$$x+y+z=4 \quad (1)$$
$$y+z=3 \quad (2)$$
$$3z=3 \quad (3)$$
$$(3) \Rightarrow \ z=1$$
$$(2) \Rightarrow \ y=3-z=2$$
$$(1) \Rightarrow x=4-y-z$$
$$x=1$$
$$\therefore \ x=1, \quad y=2, \quad z=1$$

Example 1.13

Show that the equations $x+y+z=6$, $x+2y+3z=14$, $x+4y+7z=30$ are consistent and solve them.

Solution:

The matrix equation corresponding to the given system is

$$\begin{pmatrix}1&1&1\\1&2&3\\1&4&7\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}6\\14\\30\end{pmatrix}$$
$$A \quad X = B$$

Augmented matrix $[A,B]$	Elementary Transformation
$\begin{pmatrix}1&1&1&6\\1&2&3&14\\1&4&7&30\end{pmatrix}$	
$\sim\begin{pmatrix}1&1&1&6\\0&1&2&8\\0&2&4&16\end{pmatrix}$	$R_2 \to R_2 - R_1$ $R_3 \to R_3 - R_2$
$\sim\begin{pmatrix}1&1&1&6\\0&1&2&8\\0&0&0&0\end{pmatrix}$	$R_3 \to R_3 - 2R_2$
$\rho(A)=2, \ \rho([A,B])=2$	

Obviously the last equivalent matrix is in the echelon form. It has two non-zero rows.

$$\therefore \quad \rho([A,B])=2, \ \rho(A)=2$$

$$\rho(A)=\rho([A,\ B])=2<\text{Number of unknowns.}$$

The given system is consistent and has infinitely many solutions.

The given system is equivalent to the matrix equation,

$$\begin{pmatrix}1&1&1\\0&1&2\\0&0&0\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}6\\8\\0\end{pmatrix}$$

$$x+y+z=6 \quad (1)$$
$$y+2z=8 \quad (2)$$
$$(2) \Rightarrow \ y=8-2z,$$
$$(1) \Rightarrow \ x=6-y-z=6-(8-2z)-z=z-2$$

Let us take $z=k, k\in R$, we get $x=k-2, y=8-2k$, Thus by giving different values for k we get different solutions. Hence the given system has infinitely many solutions.

Example 1.14

Show that the equations are inconsistent:

$x-4y+7z=14$, $3x+8y-2z=13$,
$7x-8y+26z=5$

Solution:

The matrix equation corresponding to the given system is

$$\begin{pmatrix}1&-4&7\\3&8&-2\\7&-8&26\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}14\\13\\5\end{pmatrix}$$
$$A \quad X = B$$

Augmented matrix $[A,B]$	Elementary Transformation
$\begin{pmatrix}1&-4&7&14\\3&8&-2&13\\7&-8&26&5\end{pmatrix}$	

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